

DRY-RUN GUIDE | left\_rotate.py  
Source: Python\4th\_belt\left\_rotate.py

#### CORE BEHAVIOR

Rotates an array left; leading values wrap around to the end.

#### REPRESENTATIVE INPUT

[1,2,3,4,5], k=2.

#### STEP TRACE

Remove prefix [1,2], retain [3,4,5], append the prefix.

#### EXPECTED OUTPUT / RETURN

[3,4,5,1,2]

#### ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\4th\_belt\left\_rotate.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Rotate the array left by one position.
002 | # Add your solution here.
003 |
004 | from typing import List
005 |
006 | def left_rotate(arr: List[int], d: int) -> None:
007 |     n = len(arr)
008 |     if n == 0:
009 |         return
010 |     d = d % n
011 |
012 |     # Helper function to reverse a segment in-place using pointers
013 |     def reverse_segment(start: int, end: int) -> None:
014 |         while start < end:
015 |             arr[start], arr[end] = arr[end], arr[start]
016 |             start += 1
017 |             end -= 1
018 |
019 |     # 1. Reverse the first 'd' elements
020 |     reverse_segment(0, d - 1)
021 |     # 2. Reverse the remaining 'n - d' elements
022 |     reverse_segment(d, n - 1)
023 |     # 3. Reverse the entire array
024 |     reverse_segment(0, n - 1)
025 |
```

#### DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.